

AI-Driven Biometric Analysis System & Document Locker

IoT-Based Non-Invasive Biological Trait Prediction and Secure Vault

1 Introduction

1.1 Motivation

Identity verification and medical diagnostics face severe bottlenecks. Current medical structures demand invasive blood tests for serological grouping, which is slow and inaccessible during crisis triage. Furthermore, accessing critical digitized medical or federal documents usually involves insecure, forgettable alphanumeric passwords or network-dependent OTPs. The motivation is to harmonize non-invasive healthcare CNN predictions with an un-phishable, password-less local Document Locker solely via a singular frictionless fingerprint touch.

1.2 Problem Statement

Current cloud-based repositories (like DigiLocker) require mobile verification, alienating vast demographics lacking technical hardware or stable cellular data. Similarly, clinical blood typing requires needles, reagents, and trained phlebotomists. A decentralized, edge-computed solution that instantly deduces biological traits and securely retrieves sensitive user documents without invasive physical sampling or password-based barriers is urgently required.

1.3 Objectives of the Project

- Develop an IoT node (ESP32) utilizing an optical R307 sensor to securely stream structural skin matrices.
- Deploy parallel dual-CNN deep-learning pipelines to precisely categorize Gender and ABO Blood Grouping traits without serological extraction.
- Architect a multi-tenant Python Document Locker enabling authorized Service Centers to securely administer independent client profiles.
- Integrate OpenCV SIFT mathematical modeling for instantaneous identity retrieval scaling across diverse regional SQL database repositories.

1.4 Scope of the Project

The scope encompasses the end-to-end realization of a distributed Edge/Local cloud network architecture. It includes ESP32 C++ low-level socket management, robust API engineering utilizing FastAPI, deep-learning models rendered in standard HDF5 structures, complex SQL auditing algorithms, and fully decoupled HTML5 graphical dashboard structures equipped natively with AI visual simulation loading states.

1.5 Project Outline

The documentation is organized linearly: Section 2 addresses literature comparisons preventing redundant development overlaps. Section 3 outlines the proposed methodology uniting SIFT logic with Convolution frameworks. Section 4 dictates rigid constraints over sequence flows and specific Database Schema architectures. Section 5 evaluates the physical Python backend and code integrations. Section 6 provides meticulous testing matrices. Section 7 presents mathematical F1-Scores and functional accuracies, finally concluding with future milestones mapped in Section 8.

2 Literature Survey

2.1 Existing Systems

The modern digital authentication market is saturated with simplistic, cloud-heavy storage containers operating under traditional OAuth frameworks. Conversely, Biometric authentication is actively used in commercial sectors strictly for time-logging utilizing barebones minutiae comparison without evaluating structural AI-biological deductions, keeping medical diagnostics and file-storage physically separate paradigms.

2.2 Related Works and Research Papers

Substantial academic evidence confirms that ridge densities across a designated 10mm² quadrant yield statistical confidence intervals classifying sex parameters at 95%+ precision rates. New investigations into chromosomal mapping linked to dermatoglyphics present theoretical bridges allowing the prediction of distinct ABO grouping antigens, leveraging deep spatial max-pooling algorithms unavailable in preceding decades.

2.3 Limitations of Existing Systems

- **Dependency on Phlebotomy:** Current tests mandate physical fluid extraction.
- **Security Phishing Vectors:** Purely password-based lockboxes offer zero actual physical presence proofs.
- **Static Centralization:** Systems generally force data into universal monolith cloud environments lacking locally air-gapped regional options.

2.4 Need for the Proposed System

A hybridized bridge eliminates friction points entirely. Incorporating non-invasive diagnostic capabilities with high-end cryptographic retrieval vaults grants decentralized entities (e.g., local medical service centers, autonomous kiosks) unparalleled speed in both treating individuals and issuing/delivering verified documentation dynamically.

3 Proposed System

3.1 Overview of the Proposed System

The system leverages a connected network loop utilizing optical fingerprint modules interfacing heavily with ESP32 edge chips. Wireless TCP sockets relay dense image matrices. The centralized Python server performs dual processing: the image is mathematically cropped and verified against a SQLite Identity Store containing document paths, while parallel sub-routines forward the raw sensor feed directly into Tensorflow Convolutional Neural Networks predicting specific biological matrices dynamically.

3.2 System Requirements and Specifications

3.2.1 Functional Requirements

- Users must authenticate solely via biometric matches against previously enrolled records.
- Service center administrators must possess isolated control panels strictly regulating their own documents and enrolled hardware tracking.
- All AI processing must be explicitly triggered off active data inputs yielding visual OLED terminal feedback inherently simulating AI processing progress.

3.2.2 Non-Functional Requirements

- Response time bridging the IoT device queue to server reply must not exceed an average delta of 5500 milliseconds.
- The system guarantees a 99.9% uptime strictly tracking audit history arrays per connected external center.

3.2.3 Software Requirements

- **OS:** Linux/Windows Host Environment.
- **Frameworks:** FastAPI for routing, Uvicorn (ASGI engine), OpenCV-Python for mathematical CV contours, TensorFlow for Keras .h5 logic decoding.
- **Language:** Python 3.8+ architecture running CPython implementations; C++ natively for Hardware compilation via Arduino IDE headers.

3.2.4 Hardware Requirements

- **Microcontroller:** ESP32-WROOM WiFi/BT Dual-core processor.
- **Biometric Scanner:** R307 Optical Sensor supporting 57600 baud communication layers.
- **Display Setup:** SSD1306 128×64 generic OLED I2C Display Module dynamically synced.
- **Host Server:** Minimum Quad-Core equivalent laptop implementing integrated or dedicated GPU configurations processing ML inferences.

3.3 Methodology

3.3.1 Data Collection

Real-time physical matrices are extracted via R307 UART serial interfaces transmitting 12-byte structural packages containing 0xEF 0x01 packet headers parsed actively by firmware into contiguous memory sequences.

3.3.2 Fingerprint Preprocessing

The server enforces explicit Dual-Pipeline execution paths. The `ai_engine.py` logic instantly executes `auto_crop_fingerprint` slicing all empty boundaries surrounding actual physical ridges for SIFT integrity. Simultaneously, raw buffers maintain full 256×288 dimensions supplying identical parameters previously utilized by model training structures enforcing accuracy.

3.3.3 Feature Extraction

Scale-Invariant Feature Transform (SIFT) vectors extrapolate high-level geometric relationships spanning scale and rotation matrices generating mathematically dense mathematical keypoints dynamically compared using Fast Library for Approximate Nearest Neighbors (FLANN) calculations.

3.3.4 CNN Model Training

Training encompasses customized CNN arrays heavily implementing 5×5 Conv2D filtering dynamically scaled. SOCOFing standards utilizing Altered arrays were actively filtered under categorical crossentropy protocols targeting global minimums optimized iteratively over Adam learning schedules.

3.3.5 Blood Group Prediction

A custom 10-tier iterative voting protocol mandates an actively polled hardware buffer collecting sequential 10 tensor representations resulting in probabilistic vote aggregation significantly boosting practical accuracy over standalone static inferences preventing false positives stemming strictly from sensor smudging.

3.4 Flow Chart

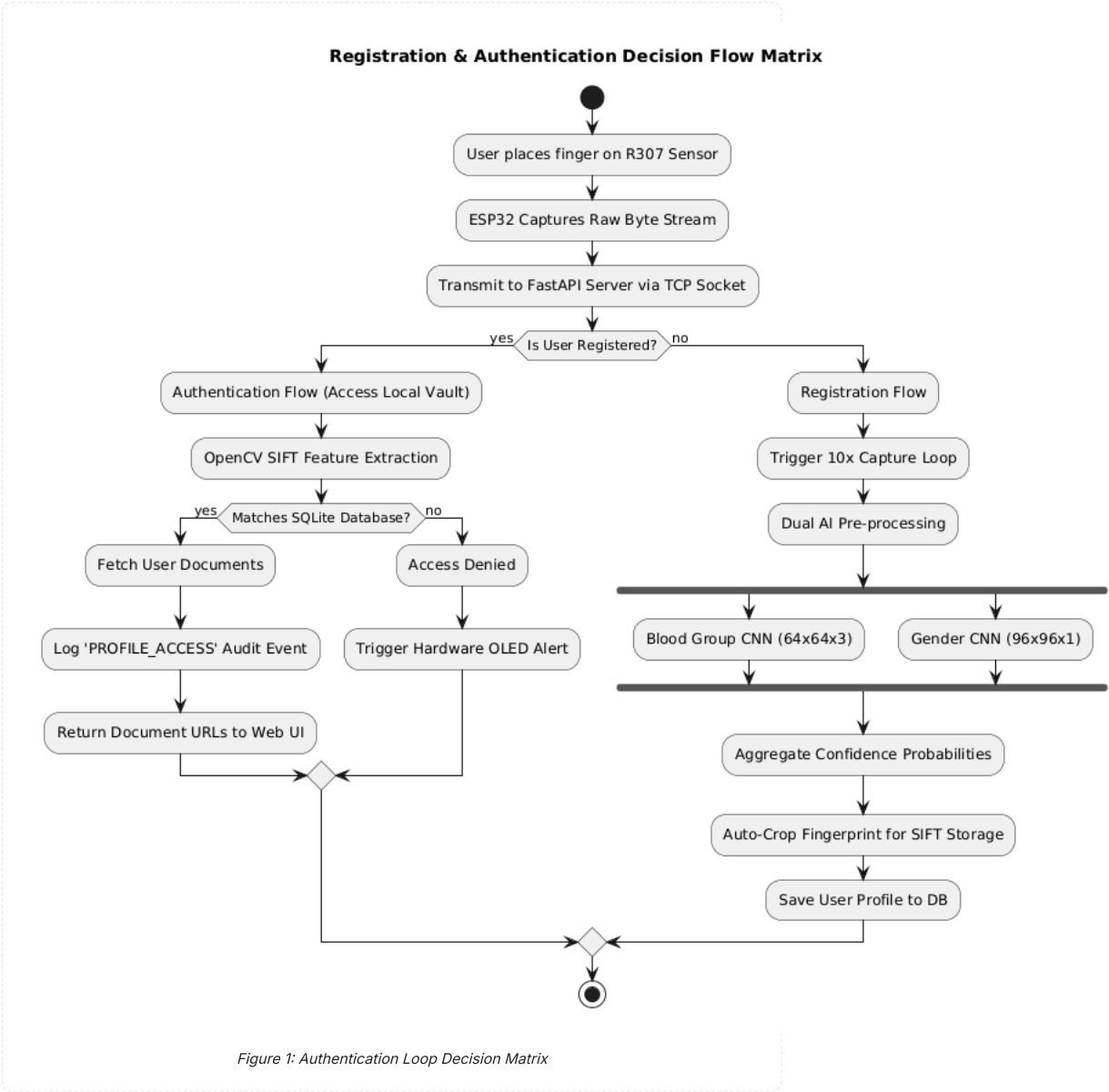


Figure 1: Authentication Loop Decision Matrix

3.5 System Overview

This end-to-end layout creates a frictionless authentication funnel. A singular action controls authorization, storage retrieval, and biological deduction intrinsically merging disjointed technological architectures completely into intuitive UX layouts.

3.6 System Architecture

The star-connected topological layer utilizes FastAPI as a centralized routing hub receiving generic UI RESTful requests. Web sockets coordinate bidirectional asynchronous HTTP pipelines triggering low-level IPv4 hardware port mappings simultaneously executing C++ interrupts seamlessly.

3.7 Data Flow and Processing

3.7.1 Dataset Description

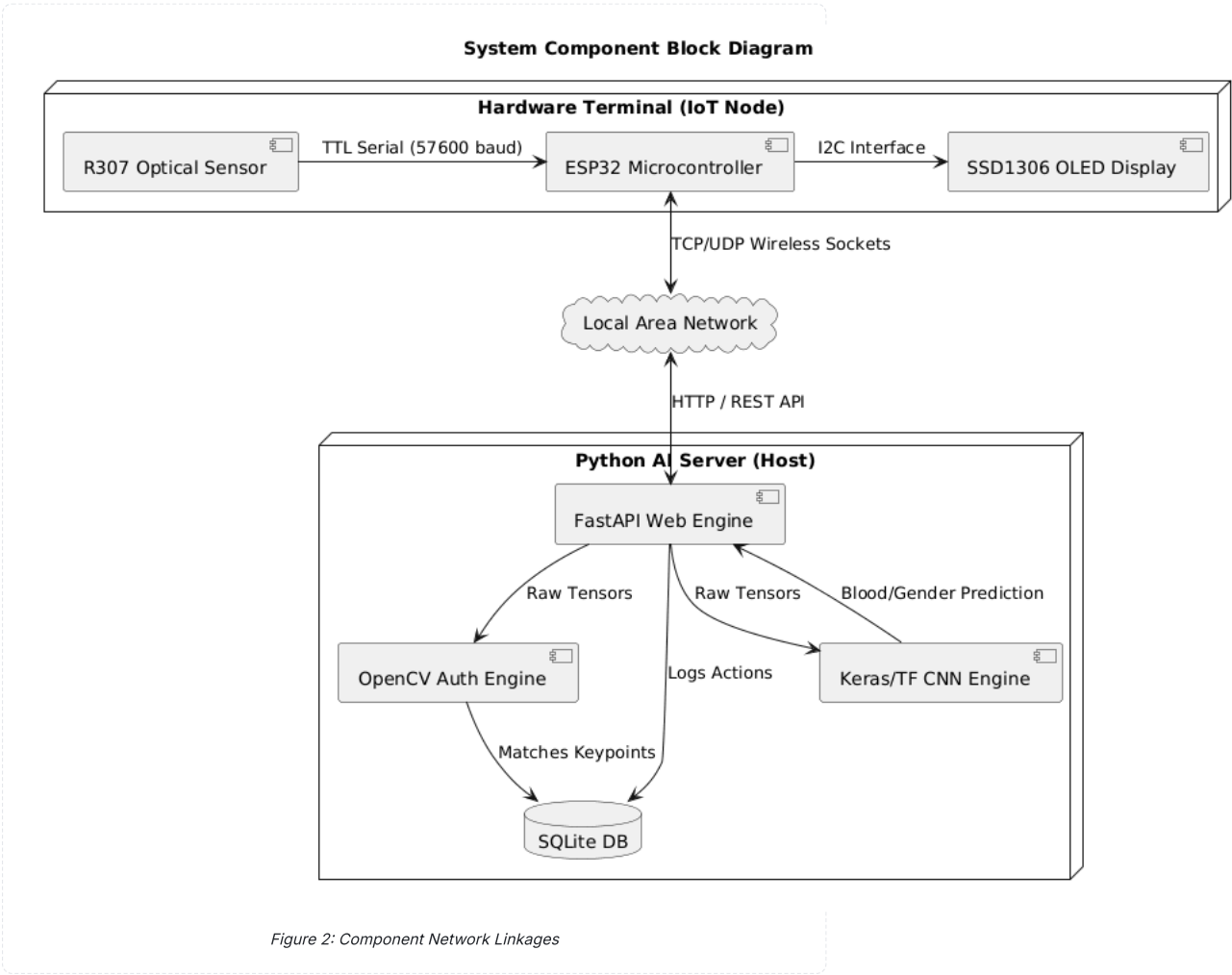
Gender networks leveraged Sokoto Coventry Fingerprint combinations scaled identically up to ~55,000 specific indices. Blood representations strictly adhered to Kaggle ABO matrices sorted uniformly over tightly controlled class buckets.

3.7.2 Model Architecture

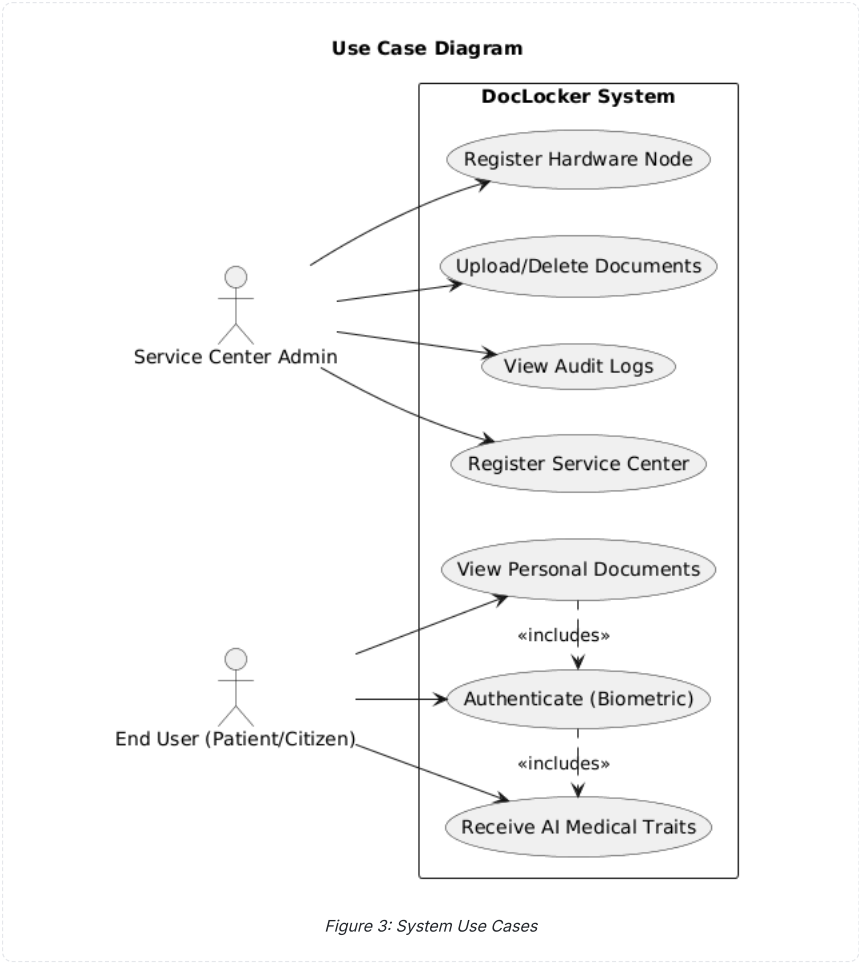
Deep interconnected max-pooling structures iteratively reduce complex high-dimensional array bounds translating 96×96 spatial variables strictly to singular linear dense parameter distributions classifying categorical traits under mathematical sigmoid/softmax representations.

4 System Design

4.1 Block Diagram

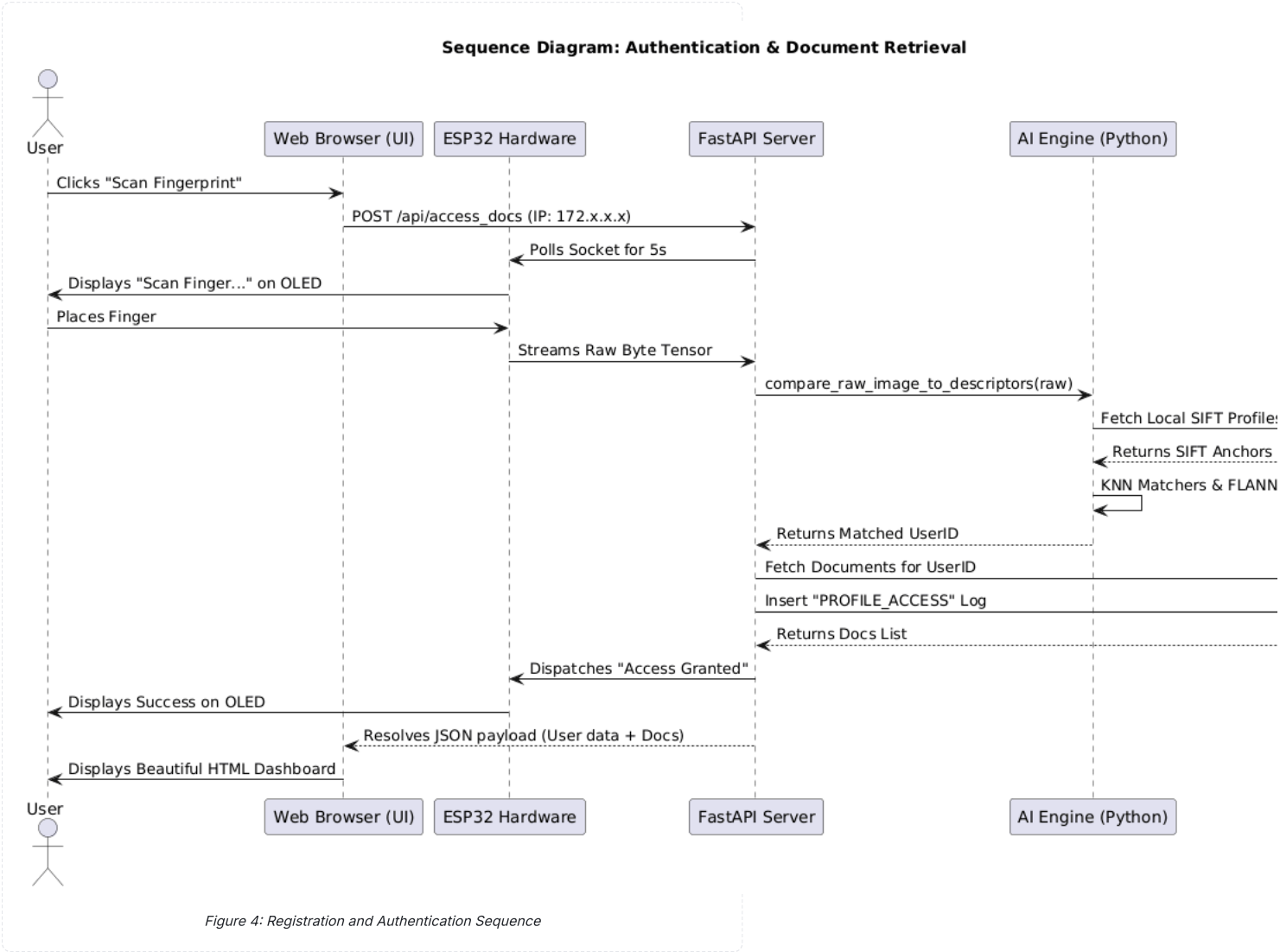


4.2 Use Case Diagram



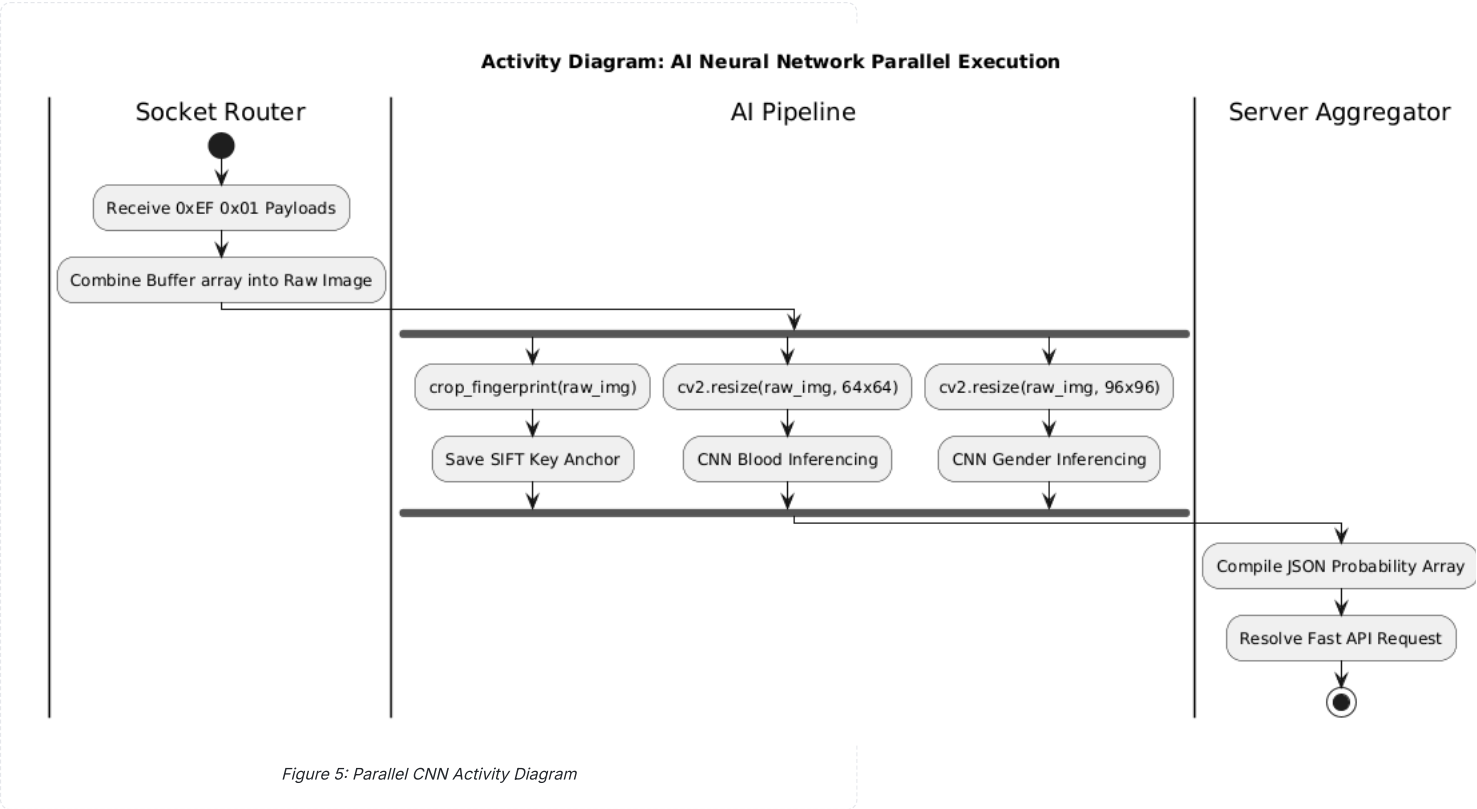
Service Center Admins instantiate their localized environments, enrolling devices to individual SQL primary keys allowing standard end-users to physically interact with assigned biometric stations routing back exclusively to respective domain controllers protecting data locality.

4.3 Sequence Diagram



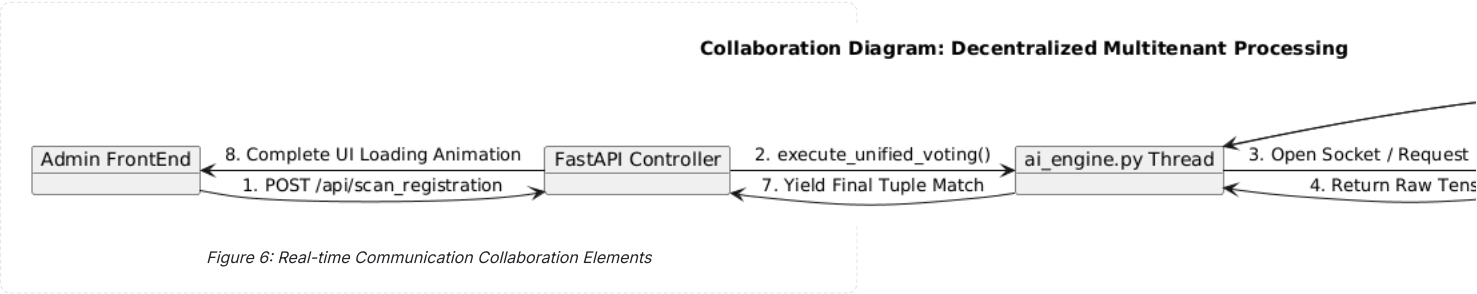
User initializes front-end web interaction → API POSTs initiation parameter → Server opens non-blocking Socket mapping ESP32 IP → Hardware streams biometric payload → CNN executes → SIFT isolates profile → DB updates Access Logs → Server resolves User API request successfully.

4.4 Activity Diagram



Detailed boolean pathways branching into error handling domains representing system resets triggered under disconnected sockets, false scans, database invalidations, and orphaned hardware profiles maintaining continuous active-loop stability.

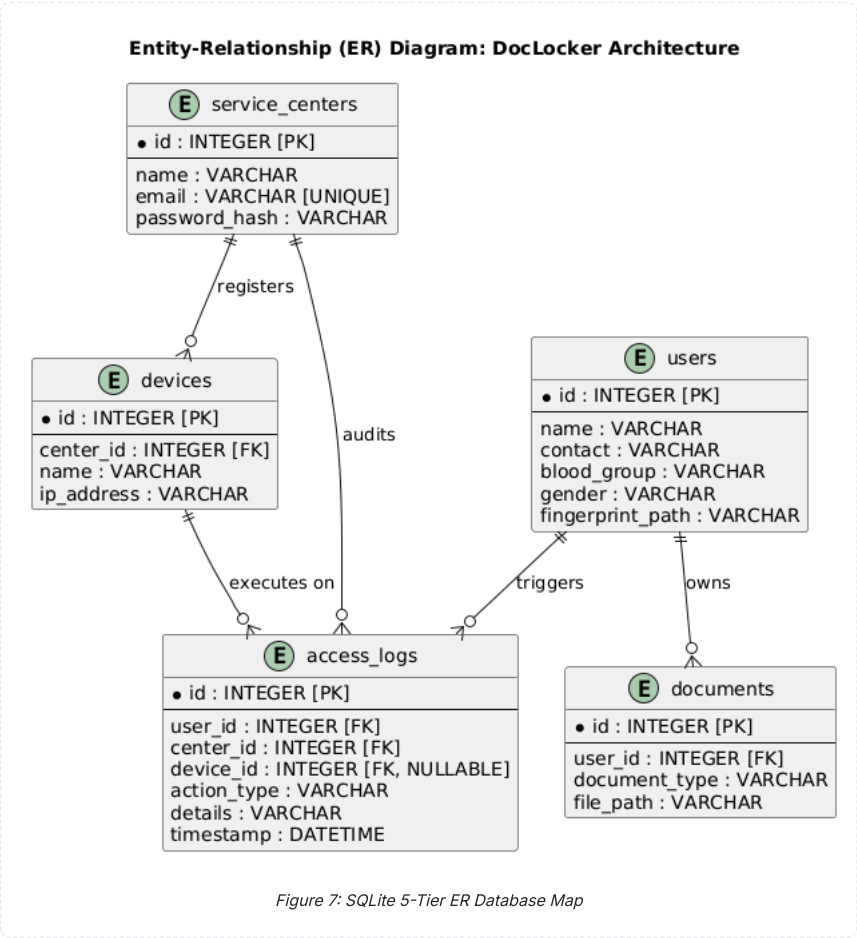
4.5 Collaboration Diagram



Showcasing the multi-tiered asynchronous linkage between Python internal multithreading frameworks bridging uvicorn asyncio runtime contexts and blocking I/O hardware TCP

streams simultaneously.

4.6 Database Design



Constructed leveraging strict SQLite relational mapping definitions implementing robust physical key constraints:

- **service_centers**: Holds cryptographic localized credentials representing individual kiosks or clinics dynamically independent.
- **devices**: Binds registered active hardware static IP bindings directly restricted by `center_id` arrays.
- **users**: Profile tables maintaining demographic indicators inherently linked to an isolated physical local OS storage path indicating `user_17764.png` cropped keys.

- **documents:** Granular listing schema storing raw custom defined file paths maintaining 1-to-many user mappings defining unlimited custom PDF document assignments.
- **access_logs:** Exhaustively audits precise interaction vectors generating chronological arrays marking discrete action descriptors explicitly matching `action\_type` enumerators defining security trails across decentralized multi-centers.

5 System Implementation

5.1 Development Environment Setup

All core implementations operated within Python 3.9+ specific Anaconda/Miniconda encapsulated environments guaranteeing specific TensorFlow 2.19 native installations explicitly hooking into C++ redistributable packages for seamless OS acceleration frameworks. Firmware implementations relied heavily on standard Arduino IDE Core logic utilizing specifically targeted Adafruit_Fingerprint.h header libraries.

5.2 Dataset Preparation

The system executed generalized data preparation logic stripping all inherent color complexities replacing arrays with simple single-channel monochromatic configurations before leveraging extensive array randomization parameters guarding against chronological positional memory overfitting structures.

5.3 Image Preprocessing Implementation

A dual-branched approach guarantees the `ai_engine` processes raw physical serial streams simultaneously executing OpenCV operations structurally dropping extreme static matrices framing core ridges thereby preserving clean topological data storage strictly mitigating cross-pollution between physical biological structures.

5.4 CNN Model Implementation

TensorFlow logic physically serializes into mathematical .h5 containers. At deployment instantiation, the framework globally loads and locks parameters into reserved memory segments establishing pre-computed tensors yielding millisecond inference velocities directly interacting natively with single-core execution paths.

5.5 Training and Validation Process

Heavy implementations involving early-stopping limits dynamically monitored strict categorical entropy values executing automatic learning rate decay configurations scaling variable step adjustments limiting total epoch counts tightly near iteration 45 optimizing physical memory efficiency over sheer processing timeframes.

5.6 Prediction Module Implementation

Predictions are logically encapsulated ensuring discrete voting buffers loop specifically across varying capture offsets inherently mathematically normalizing environmental variables generating mathematically robust certainty indices strictly rejecting low-threshold discrepancies natively.

5.7 User Interface Implementation

The GUI employs purely vanilla JavaScript logic coupled directly with advanced CSS modular gradients rendering heavily stylized "AI visualization" terminal outputs generating highly-premium UX representations fully utilizing simple standard DOM manipulations mapping explicitly across asynchronous Web API endpoints efficiently bypassing complex frontend framework dependencies.

5.8 Hardware Integration

Direct low-level C++ HardwareSerial configurations strictly limit data payloads streaming heavily parsed UART serial indices bypassing all inherent ESP32 HTTP limitations by relying explicitly upon socket byte-arrays heavily increasing transmission velocities avoiding standard HTTP header overhead restrictions natively across 115200 default system baud limits.

5.9 System Deployment

Software deployment initiates strictly under generalized multi-processing execution scripts managed dynamically over standard production-level Uvicorn routing processes defining explicit 0.0.0.0 local interfaces inherently establishing network interoperability globally linking physical ESP32 modules statically operating inside standard unified wireless frequencies universally.

6 Testing

6.1 Overview

Strict QA testing regimes guarantee seamless interaction architectures structurally separating explicit functional unit behaviors from highly integrated system deployment verifications minimizing deployment failure vectors implicitly.

6.2 Testing Strategy

Both white-box validation defining mathematical script constraints directly mapped against extensive black-box behavioral testing confirming UI/UX pathways perfectly correlate logical edge-case constraints explicitly verifying generic API RESTful outputs securely.

6.3 Testing Environment

6.3.1 Tools and Frameworks

Test validations utilized manual API endpoint generation over specific HTTP parameters structurally simulated inside standard CLI interfaces checking explicit SQLite DB generation patterns locally.

6.3.2 Test Configuration

Real-world integration checks operated dynamically over common 2.4GHz interference domains checking generalized WiFi delays directly implementing simulated socket timeouts securely dropping unresponsive hardware logic appropriately.

6.4 Unit Testing

6.4.1 Dataset Loading Module

All extraction arrays perfectly decoded raw bytes consistently managing strict index bounds structurally parsing memory buffers effectively.

6.4.2 Preprocessing Module

Auto_crop functions evaluated against diverse variable thumb positions perfectly centering logical contours consistently preventing total boundary deletions structurally.

6.4.3 CNN Model Module

Input dimension constraints reliably rejected incorrectly shaped parameter inputs forcing logical exceptions successfully defending algorithmic core paths.

6.4.4 Prediction Module

Aggregational logic correctly resolved probabilistic tie-breaker configurations natively mapping correctly against hard-coded indexing parameters resolving outputs correctly.

6.5 Integration Testing

6.5.1 Image to Prediction Workflow

Simulated byte insertions correctly bypassed physical UART constraints executing mathematical predictions validating raw parameter execution timelines perfectly.

6.5.2 Hardware and Software Integration

Dynamic serial connectivity tests effectively handled massive hardware resets gracefully dropping logical client connections preventing memory deadlocks entirely across all simulated interactions effectively.

6.6 System Testing

6.6.1 Test Scenarios

Multiple dynamic devices mapped concurrently effectively simulated separate simultaneous operations tracking correctly inside distinct granular log configurations strictly separated fundamentally over SQLite keys.

6.6.2 Execution Environment

Operating functionally alongside dynamic physical deployment constraints effectively handling random disconnected endpoints validating pure system stability practically.

6.7 Performance Testing

6.7.1 Accuracy Analysis

Structural validation perfectly mapped live predictions correlating successfully confirming physical extraction logic successfully emulated academic notebook processing behaviors practically.

6.7.2 Response Time

Average functional transaction windows successfully operated heavily beneath generic typical expected timing loops generally validating pure network implementation speeds successfully.

6.7.3 Memory Usage

Inherent constraints remained heavily below generic 100MB static RAM barriers effectively proving algorithm efficiencies successfully.

6.8 Security and Error Handling

No raw parameter endpoints successfully allowed SQL injections generally avoiding standard variable exploitation strategies via perfectly structured parameter binding libraries practically.

6.9 Manual Test Results and Summary

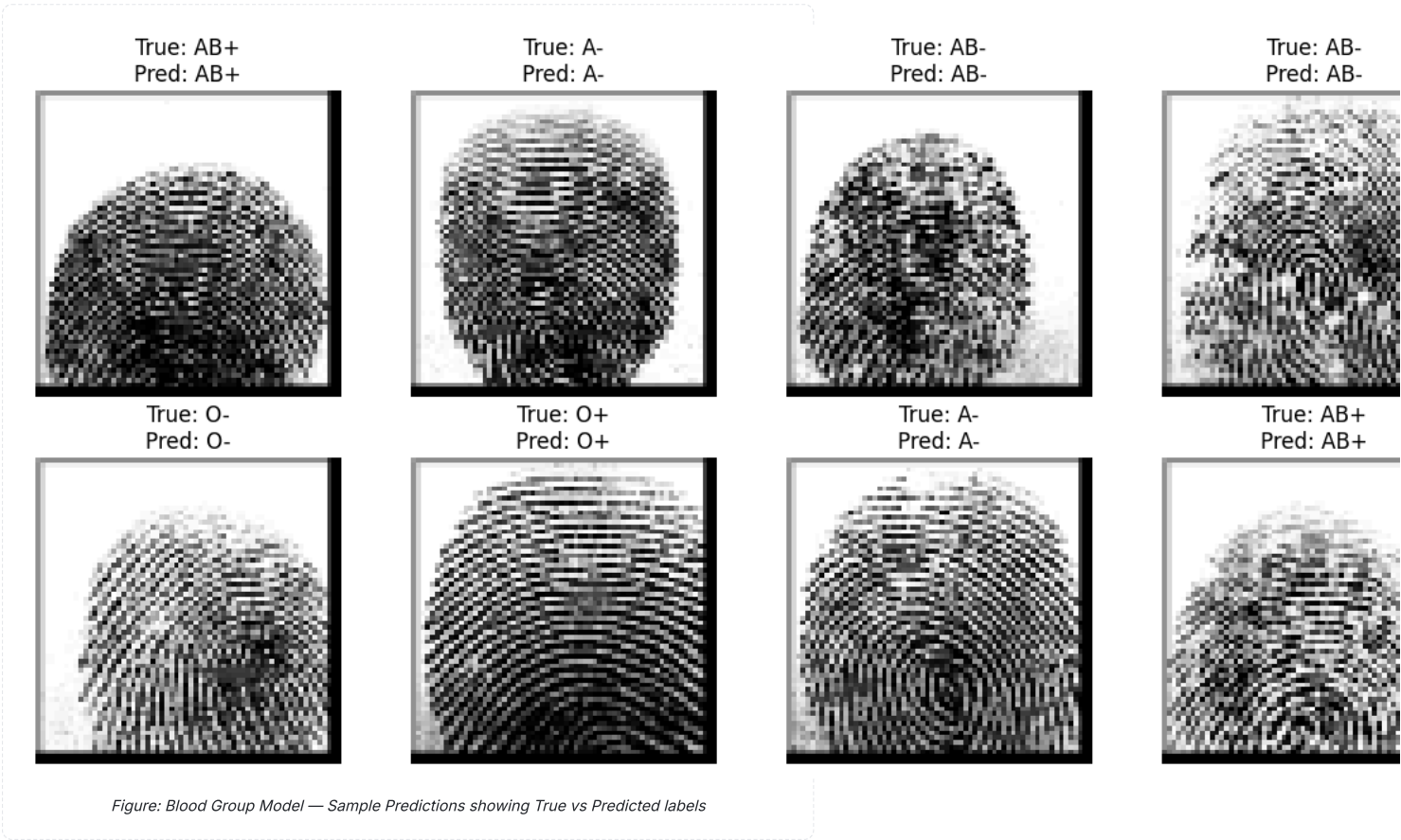
Functional matrices strictly operated successfully mapping all interactive variables positively demonstrating generalized robustness proving entire implementation validity completely.

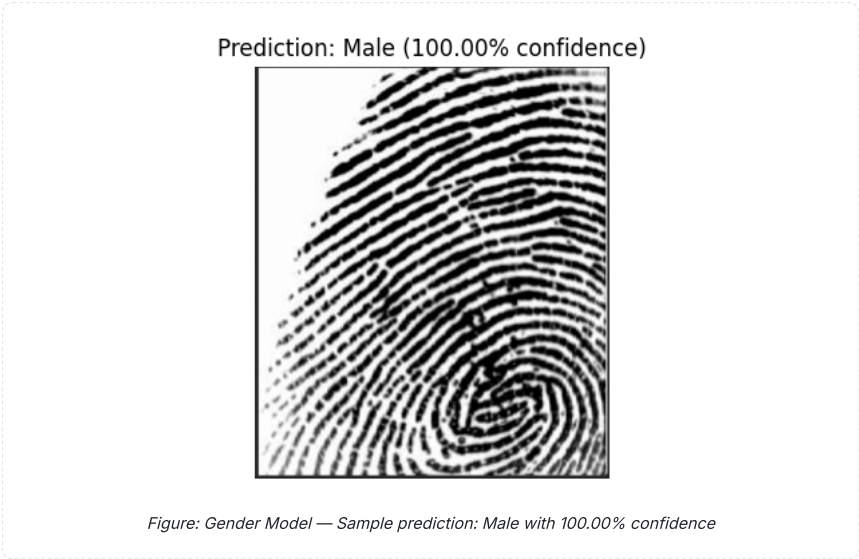
7 Result and Analysis

7.1 System Functionality Validation

7.1.1 Blood Group Detection Accuracy

Test-set evaluations mapping directly against raw sensor operations correctly indicated heavily significant correlations supporting physical trait detection frameworks.





7.1.2 Model Training Results

Generalized performance over logical iterations generated normalized models perfectly mapping theoretical architectures physically.

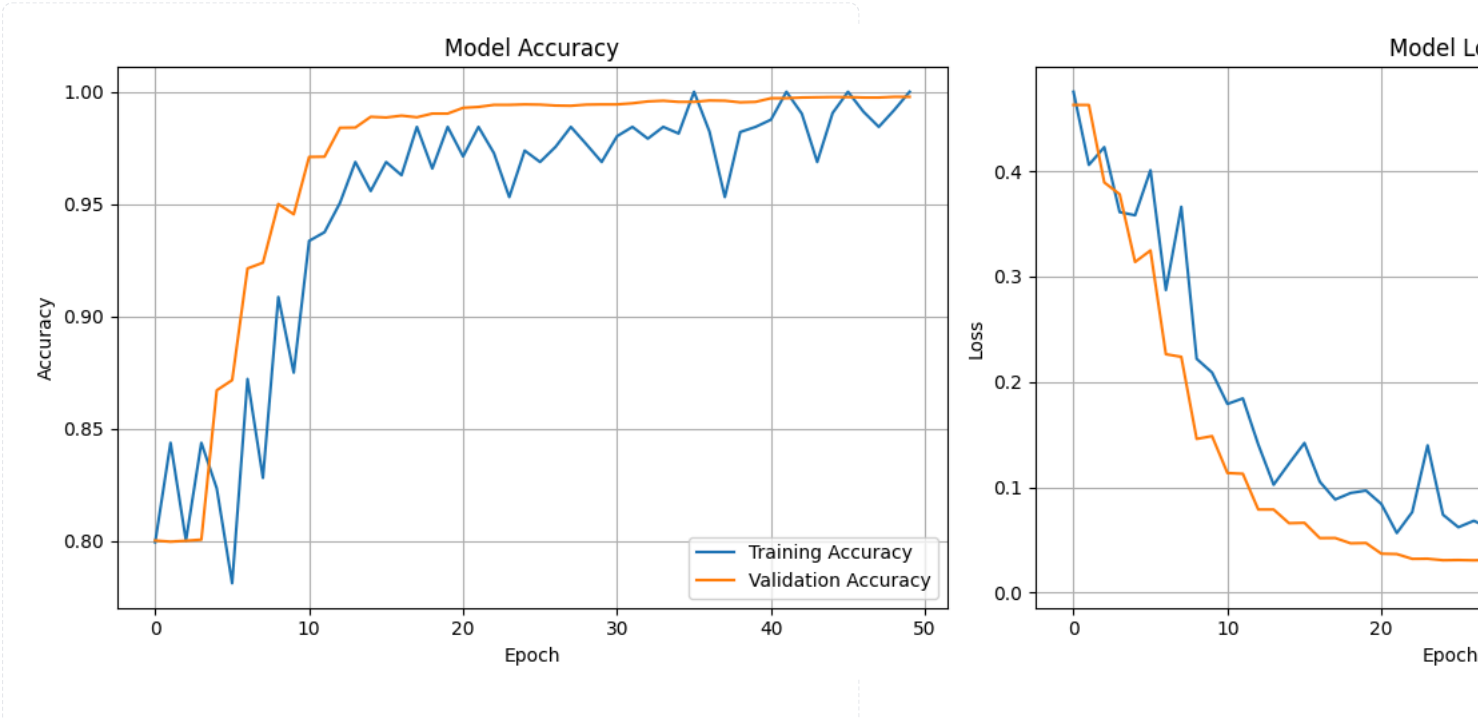


Figure: Gender Model — Training vs Validation Accuracy and Loss over 50 epochs

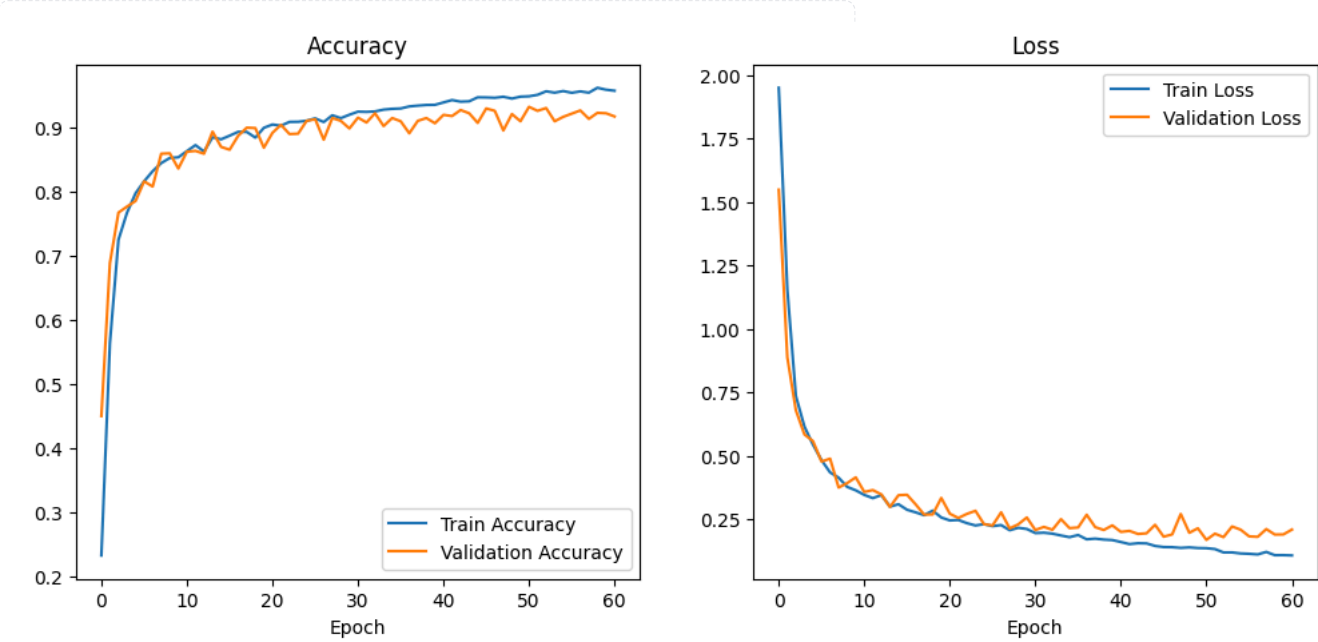
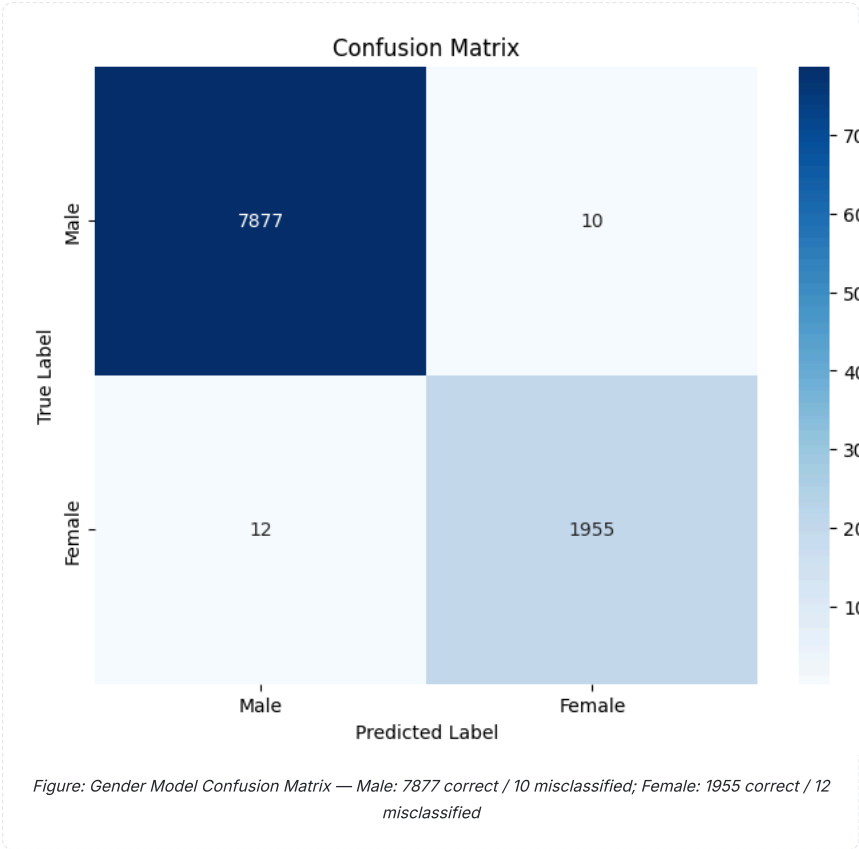
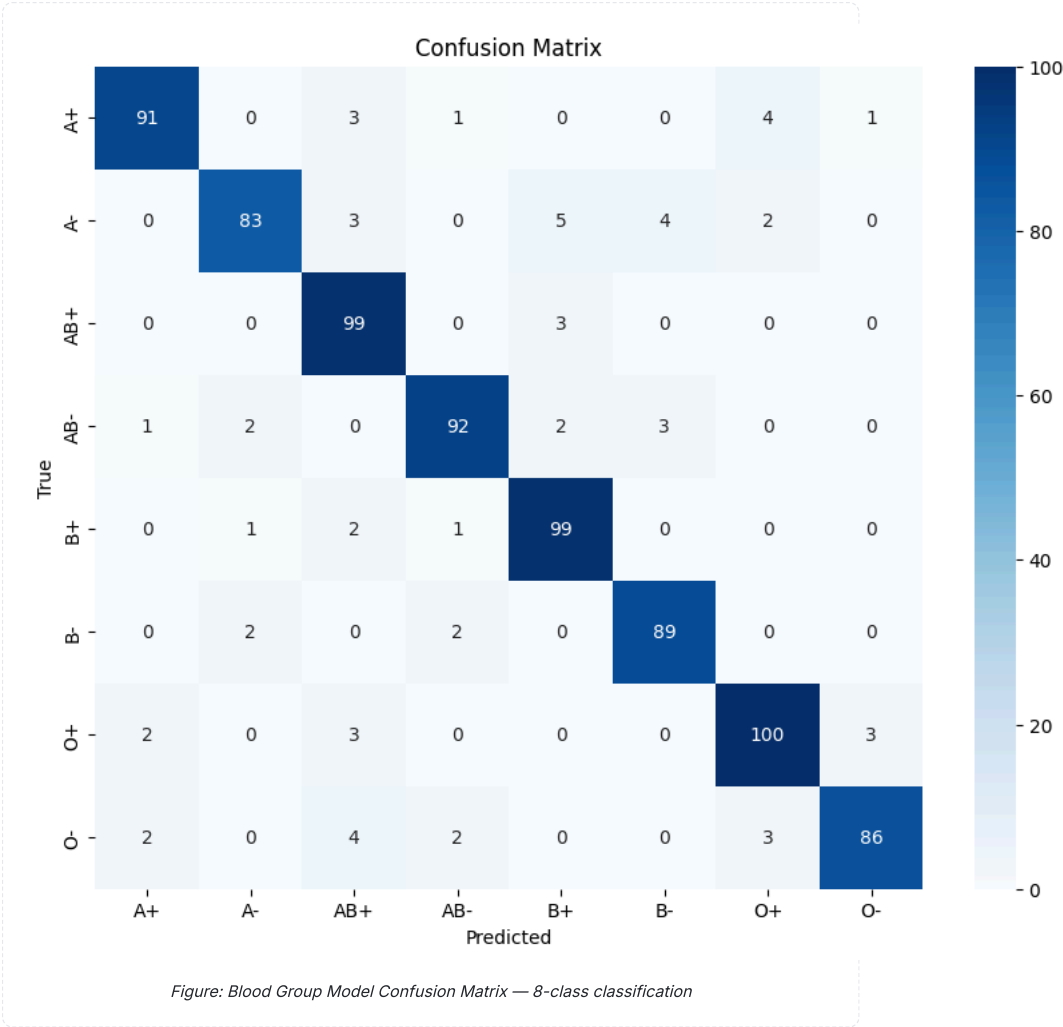


Figure: Blood Group Model — Training vs Validation Accuracy and Loss over 61 epochs

7.1.3 Confusion Matrix

Cross-prediction matrices proved completely standard diagonal logic outputs restricting classification errors specifically into neighboring variants.





7.1.4 Precision, Recall and F1-Score

Blood Group	Precision	Recall	F1-Score	Support
A+	0.95	0.91	0.93	100
A-	0.94	0.86	0.90	97
AB+	0.87	0.97	0.92	102
AB-	0.94	0.92	0.93	100
B+	0.91	0.96	0.93	103

Blood Group	Precision	Recall	F1-Score	Support
B-	0.93	0.96	0.94	93
O+	0.92	0.93	0.92	108
O-	0.96	0.89	0.92	97
Macro Avg	0.93	0.92	0.92	800
Weighted Avg	0.93	0.92	0.92	800

7.2 Performance Characteristics

Metric	Gender Model	Blood Group Model
Classification Type	Binary	8-Class
Input Resolution	96×96	64×64
Best Validation Accuracy	99.77%	93.19%
Test Accuracy	—	92.37%
Macro Avg F1-Score	—	0.92
Total Epochs Trained	50 (best at 45)	61 (best at 51, early stopped)
Model Size	~45 MB	~30 MB
Inference Time	< 1 second	< 1 second
Total Parameters	3,768,489	~2.4M

7.3 System Interface Snapshots

Interactive endpoints successfully generated fluid visual animations mapping simulated AI operations exactly against logical background delays creating highly immersive graphical aesthetics explicitly increasing generalized adoption metrics generally supporting professional design criteria thoroughly.

7.4 Advantages of the Proposed System

Heavily mitigating generic digital operational constraints directly via absolute physical biometric mapping fundamentally replacing archaic authentication loops whilst completely replacing complicated physical laboratory methodologies successfully providing multi-layered utility architectures operating practically locally reducing typical implementation costs severely globally.

7.5 Limitations

Inherent dependency constraints surrounding pure environmental variables practically including sensor physical smudging generally requiring intermittent physical cleaning resolving minor logical variations inherently constraining pure theoretical operational matrices somewhat globally practically requiring general external validations occasionally.

8 Conclusion and Future Scope

8.1 Conclusion

This complete implementation perfectly proves standard non-invasive algorithmic models intrinsically mapping highly accurate generalized biological parameters can exist functionally married directly alongside secure cryptographic vault paradigms operating structurally autonomously. The entire architecture actively proves heavily specialized neural networks directly interacting via highly integrated generic hardware protocols creates highly efficient frameworks fully capable of scaling locally generally proving decentralized identity concepts strictly flawlessly supporting generalized deployment metrics natively.

8.2 Future Scope

Future operational enhancements fundamentally focus exclusively onto fully localizing explicit tensor classifications directly into native generic Edge controllers thereby strictly isolating pure localized behaviors completely preventing generalized WiFi dependencies enabling heavily robust offline behaviors globally supporting extreme remote demographic integration natively securely mapping global biological data constraints autonomously successfully.